

Data visualisation and exploration

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Class details

- 15h: 2h*6 +3h presentation
- Grade: 30% Labs
70% Project/presentation
- For anonymous feedback: [Feedback - Google Docs](#)

Class organisation

- Lectures
- Labs during class
- Project

Syllabus

- **Session 1:** Introduction to Data Visualization & Data Sources
- **Session 2:** Good Practices & Common Pitfalls
- **Session 3:** Matplotlib & Seaborn: Visualization Essentials
- **Session 4:** Cartography & Geospatial Visualization
- **Session 5:** Case Studies: Data Journalism/ POWER BI (AIMS)
- **Session 6:** Visualizing Multidimensional Data & Dimensionality Reduction
(Work on your project)
- **Session 7:** Project presentation

Project

The project is due in jupyter notebook format + PPT presentation

You will choose a dataset and analyse it. You are free to select the data you want and answer the questions you want too. However, your notebook must include:

- **Introduction**

- What is the subject and context of the project?
- What is the source of the data?
- Who collected the data ?
- When was the data collected ?
- How was the data collected ?
- In what context/for what purpose was the data collected?
- What does the dataset contain? Describe the data, its type (continuous, categorical...)
- What is the size of the dataset?
- What is the dataset's license?

Project

- **What do you want to answer with those data?**
 - Cite your research question and your 2 subquestions
 - Do your sub questions allow you to answer your main research question ?
 - Do you have the data to answer your subquestions ?
 - What are your hypotheses?
- **FOR EACH SUB QUESTIONS**

Explain its relevance to your main question.

 - What is your methodology for analysis?
 - Produce at least one graphical representation (chart) per sub-question: - The code is clear and readable
 - The chosen graphical representation is consistent with the type of data and the analysis approach.
 - Graphic representation is complete
 - The graphical representation respects best practices
 - Graphical representation minimizes potential bias

- What analysis can you draw from the graphs? A detailed analysis is expected.

 - What are your results?
 - What are the limitations?

Project

- **Conclusions**

Summarize your main findings

- Does your analysis enable you to answer your research question? Why or why not?
- What are the limitations and biases of your methodology?
- Do you have any sources that support or contradict your conclusions?
- What would be the next steps in your analysis if you had had more time?
- Can you think of another data analysis method to answer your question? Expand.

- **Presentations**

10 min of presentation+5min questions. Group of 4: 1 sub question and 1 figure each

The grade will be defined as:

- 5pts: Overall presentation: clarity, slides
- 10pts: Work: what you have done (different for each member)
- 3pts: Answering the questions (different for each member)
- 2pts: Critical thinking